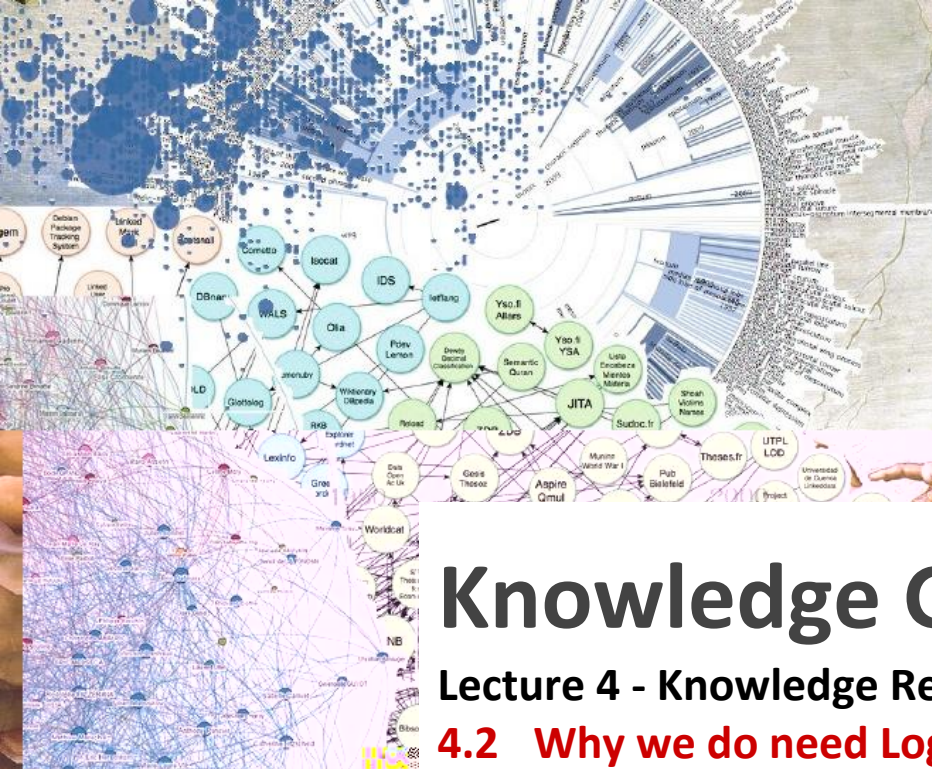




Knowledge Graphs

Lecture 4 - Knowledge Representation with Ontologies

4.2 Why we do need Logic

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Autumn 2020



FIZ Karlsruhe

Leibniz-Institut für Informationsinfrastruktur

Knowledge Graphs

Lecture 4: Knowledge Representation with Ontologies

4.1 A Brief History of Ontologies

4.2 Why we do need Logic

Excursion 4: A Brief Recap of Essential Logics

Excursion 5: Description Logics

4.3 First Steps in OWL

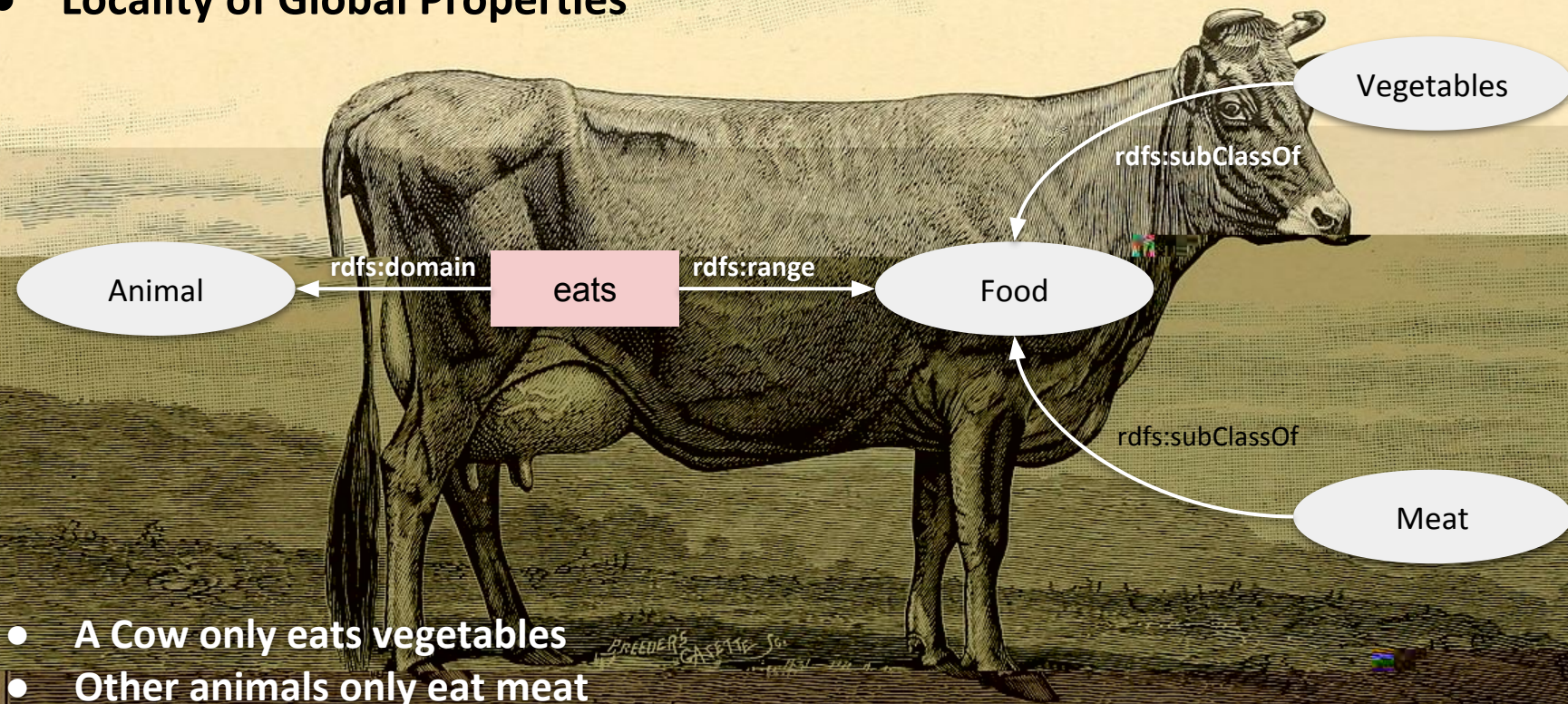
4.4 More OWL

4.5 OWL and beyond

4.6 How to Design your own Ontology

Why RDF(S) is not sufficient

- **Locality of Global Properties**

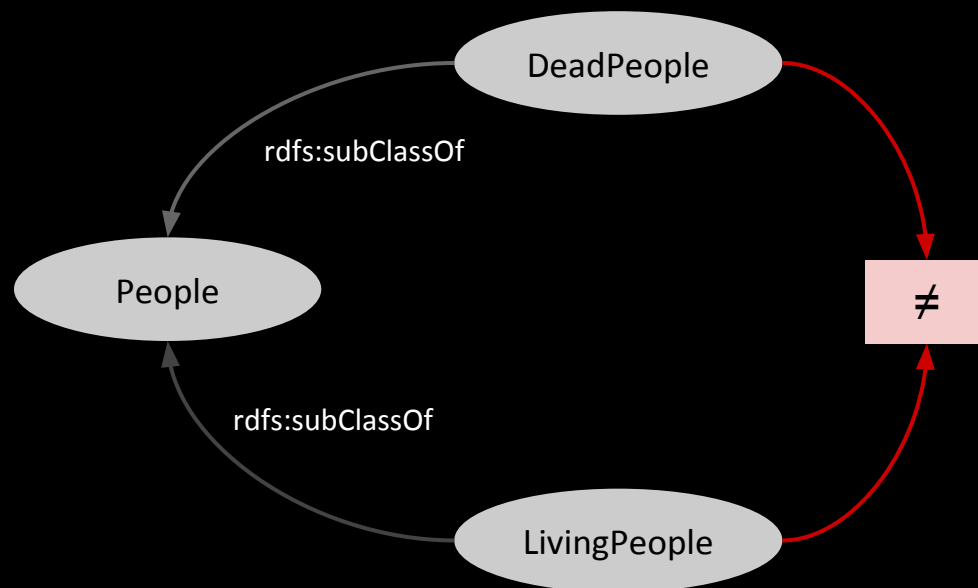


- A Cow only eats vegetables
- Other animals only eat meat

Why RDF(S) is not sufficient



- **Disjunctive Classes**

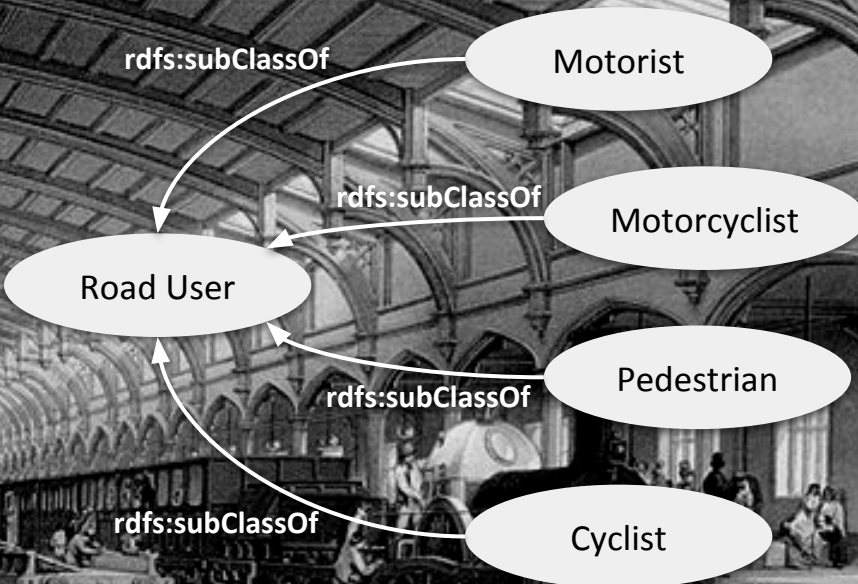


- RDFS Subclass relation cannot express disjunctive class (subclass) membership

Why RDF(S) is not sufficient

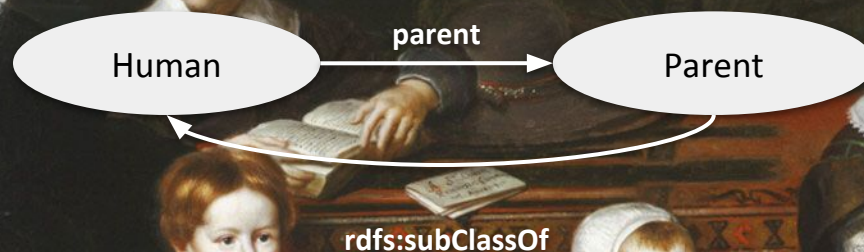
Class Combinations

- Combination of classes define a new class
- New class contains only members from given class combinations



Why RDF(S) is not sufficient

- Cardinality Constraints



- Every human (usually) has two parents

Why RDF(S) is not sufficient

Special Property Constraints

- Transitivity (e.g. „is greater than“)
- Uniqueness (e.g. „is mother of“)
- Inversiveness (e.g. „is parent of“ and „is child of“)

General Problem of RDF(S)

- RDF(S) does not have the possibility of **negation**
 - `:harald rdf:type :Vegetarian .`
 - `:harald rdf:type :NonVegetarian .`
 - ...does not automatically generate a contradiction

Why RDF(S) is not sufficient

An **ontology** is an **explicit, formal specification** of a shared conceptualization.

acc. to Thomas R. Gruber: A Translation Approach to Portable Ontology Specifications. Knowledge Acquisition, 5(2):199-220, 1993.

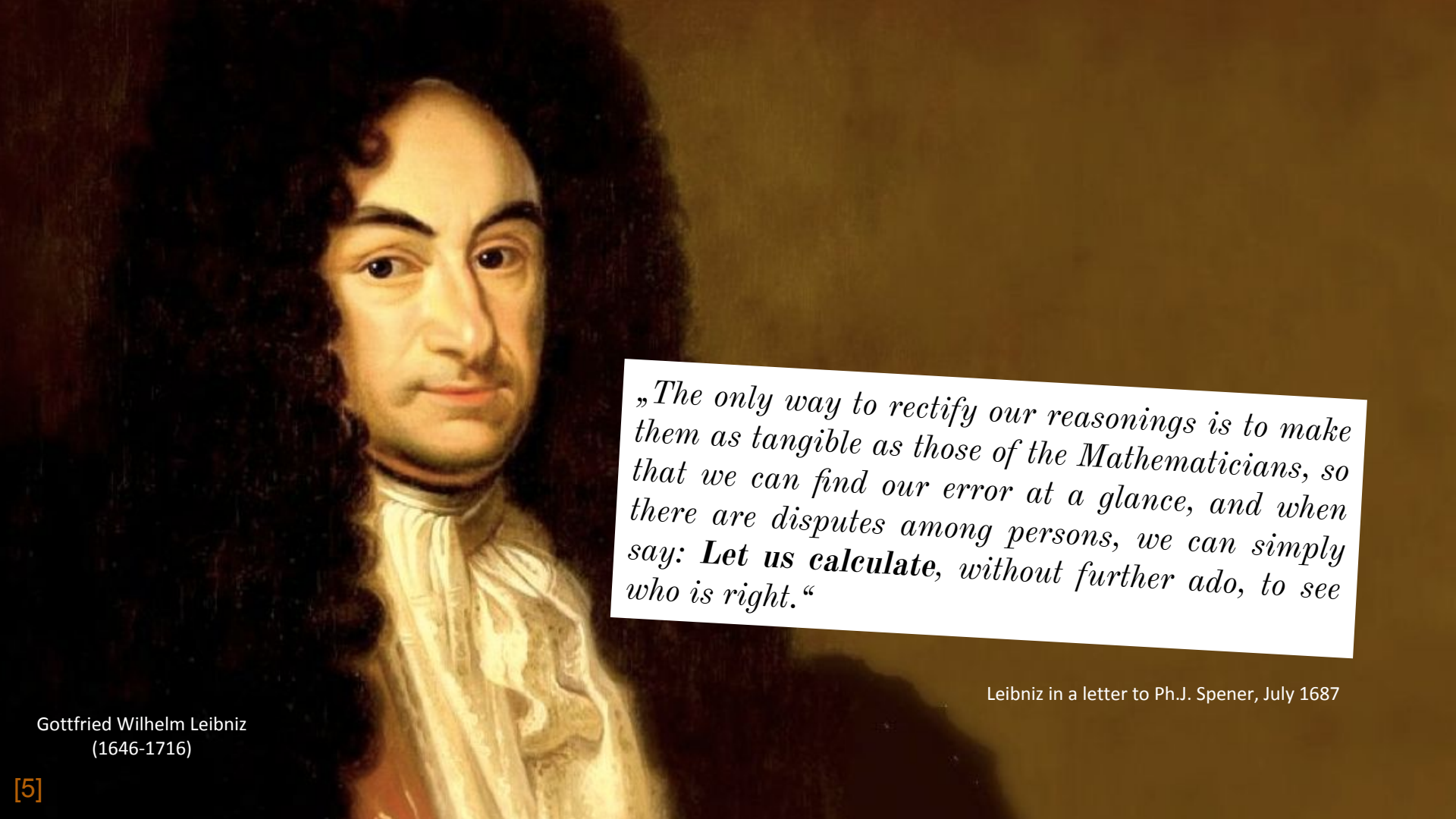
formal (machine readable) semantics

mathematical logic

The Foundations of Logic

- Definition (for our lecture):
Logic is the study of how to make formal correct deductions and inferences.

- Why “formal logic”? $\Rightarrow$ to enable automation



*„The only way to rectify our reasonings is to make them as tangible as those of the Mathematicians, so that we can find our error at a glance, and when there are disputes among persons, we can simply say: **Let us calculate**, without further ado, to see who is right.“*

Gottfried Wilhelm Leibniz
(1646-1716)

Leibniz in a letter to Ph.J. Spener, July 1687

The Foundations of Logic

- **Syntax:** *symbols without meaning*
defines rules, how to construct well-formed and valid sequences of symbols (strings)
- **Semantic:** *meaning of syntax*
defines rules how the meaning of complex sequences of symbols can be derived from atomic sequences of symbols.

Syntax

```
If (i<0) then display ("negative account!")
```

assignment of meaning

print the message "negative account!", if the account balance is negative

The Importance of Semantics

- Why should I care about semantics?

Well, from a philosophical POV, we need to specify the relationship between statements in the logic and the existential phenomena they describe.



Bertrand Russell
(1872-1970) [6]

The Importance of Semantics

- Why should I care about semantics?

Well, from a philosophical POV, we need to specify the relationship between statements in the logic and the existential phenomena they describe.



Bertrand Russell
(1872-1970) [6]

- That's OK, but I don't get paid for philosophy.

From a practical POV, in order to specify, build and test (ontology-based) tools/systems we need to precisely define relationships (like entailment) between logical statements – this defines the intended behaviour of tools/systems.

Cordial thanks to [Ian Horrocks' seminar slides!!](#)

Variants of Semantics

- e.g. programming languages

Syntax

```
FUNCTION  
f(n:natural):natural;  
BEGIN  
    IF n=0 THEN f:=1  
    ELSE f:=n*f(n-1);  
END;
```

computation of the factorial

intentional semantics

- „the meaning intended by the user“
- restricts the set of all possible models (meanings) to the meaning intended by the (human) user

Variants of Semantics

- e.g. programming languages

Syntax

```
FUNCTION  
f(n:natural):natural;  
BEGIN  
    IF n=0 THEN f:=1  
    ELSE f:=n*f(n-1);  
END;
```

computation of the factorial

$f:n \rightarrow n!$

formal semantics

*aims to express the meaning of symbol sequences (programs) in a **formal language**, in a way that assertions over the symbol sequences (programs) can be proven by the application of deduction rules.*

Variants of Semantics

- e.g. programming languages

Syntax

```
FUNCTION  
f(n:natural):natural;  
BEGIN  
    IF n=0 THEN f:=1  
    ELSE f:=n*f(n-1);  
END;
```

computation of the factorial

$f:n \rightarrow n!$

behaviour of the program at execution

procedural semantics

the meaning of a language expression (program) is the procedure that takes place internally, whenever the expression does occur.

Semantics and Mathematical Logic

- How do I define the semantics of a mathematical logic?

*In mathematical logic we define the semantics in terms of **models** (a model theory). A model is supposed to be an analogue of (part of) the world being modeled.*



Bertrand Russell
(1872-1970) [6]

Model-theoretic Semantics



Alfred Tarski
(1901-1983)

[7]

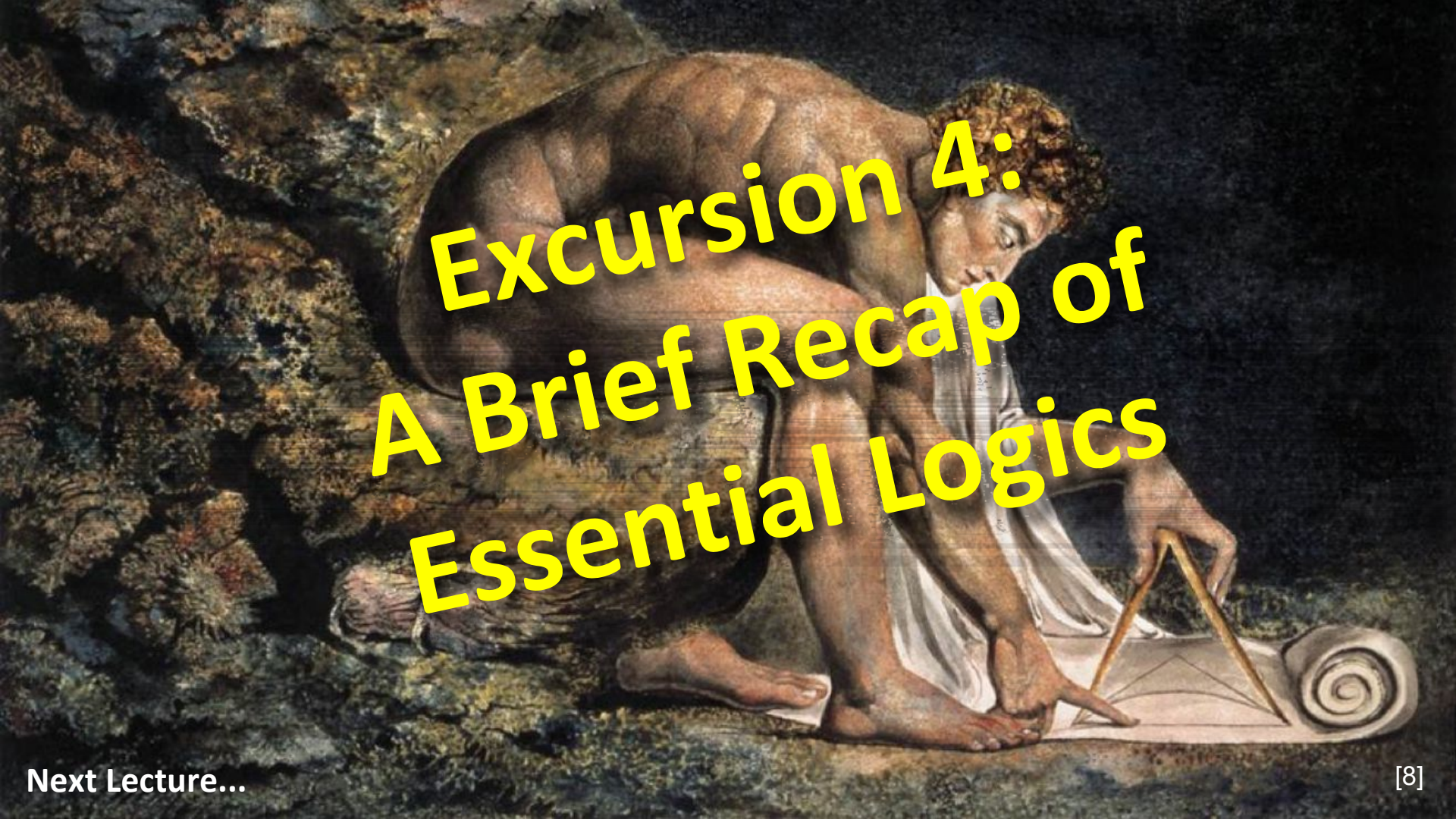
- **Model-theoretic semantics** performs the semantic interpretation of artificial and natural languages by *„identifying meaning with an exact and formally defined interpretation with a model“*
- = **formal Interpretation with a model**
- e.g. model-theoretic semantics of **propositional logic**
 - assignment of truth values *„true“* and *„false“* to atomic assertions and
 - description of logical connectives with *truth tables*

Model-theoretic Semantics

- Any logic $L := (S, \models)$ consists of
 - a **set of statements** S and
 - an **entailment relation** $\models$
- Let $\Phi \subseteq S$ and $\varphi \in S$: $\Phi \models \varphi$

*„ φ is a logical consequence of Φ “ or
„from the assertions of Φ follows the assertion φ “*

- If for 2 assertions $\varphi, \psi \in S$
both $\{\varphi\} \models \psi$ and $\{\psi\} \models \varphi$,
then both assertions φ and ψ are logically equivalent: $\varphi \equiv \psi$



Excursion 4: A Brief Recap of Essential Logics

Next Lecture...

Picture References:

- [1] Sanders, James Harvey, Jersey cow Mary Anne of St. Lambert (1887) [Public Domain]
[https://commons.wikimedia.org/wiki/File:The_breeds_of_live_stock_and_the_principles_of_heredity_\(1887\)_14781069142.jpg](https://commons.wikimedia.org/wiki/File:The_breeds_of_live_stock_and_the_principles_of_heredity_(1887)_14781069142.jpg)
- [2] Lucas Cranach the Elder, Adam and Eva , 1527 [Public Domain]
[https://commons.wikimedia.org/wiki/File:Lucas_Cranach_\(I\)_-_Adam_and_Eve_\(1527,_Pushkin_Museum\).jpg](https://commons.wikimedia.org/wiki/File:Lucas_Cranach_(I)_-_Adam_and_Eve_(1527,_Pushkin_Museum).jpg)
- [3] John Cooke Bourne, Engraving print of the inside of Isambard Kingdom Brunel's train-shed at Bristol Temple Meads railway station in the UK, 1843 [Public Domain]
https://de.m.wikipedia.org/wiki/Datei:Bristol_Temple_Meads_railway_station_train-shed_engraving.jpg
- [4] Cornelis de Vos, Portrait of Anthony Reyniers and His Family, 1631, [Public Domain]
https://commons.wikimedia.org/wiki/File:Cornelis_de_Vos_-_Portrait_of_Anthony_Reyniers_and_His_Family.jpg
- [5] Andreas Scheits, A portrait of Gottfried Wilhelm von Leibniz, 1703 [Public Domain]
https://commons.wikimedia.org/wiki/File:Leibniz_Hannover.jpg?uselang=de
- [6] James Francis Horrabin, "Bertrand Russell". The Masses: 37, 1917, [Public Domain]
https://commons.wikimedia.org/wiki/File:Bertrand_Russell,_by_J._F._Horrabin.jpg
- [7] George Bergman, Alfred Tarski, The Oberwolfach photo collection, 1968, [GFDL]
<https://commons.wikimedia.org/wiki/File:AlfredTarski1968.jpeg>
- [8] William Blake's Newton, 1795 [Public Domain]
<https://commons.wikimedia.org/wiki/File:Newton-WilliamBlake.jpg>